

4339 - ASSET PRICING AND INVESTMENTS

ASSIGNMENT 3

Exercises (pencil and paper)

1. **Factors and Arbitrage** You believe stock returns have a factor structure with two factors f_1 and f_2 . The risk-free rate is 3% and there are two well-diversified portfolios, A and B , with the following properties:

Portfolio	β_1	β_2	$E[r]$
A	1	0.5	7.5%
B	0.2	2	13.4%

- (a) Find the price of risk for each factor under no arbitrage.
- (b) Suppose there is another well-diversified portfolio C with factor loading $(\beta_{C1}, \beta_{C2}) = (0.5, 1.0)$ and expected return $E[r_C] = 9.5\%$. Is there an arbitrage opportunity? If yes, explain how to exploit this arbitrage.
2. **Performance Measures** Consider the following two portfolios

Portfolio	$E[r^e]$	σ
Market	10%	25%
P	15%	35%

- (a) Construct a portfolio P^* that has the same Sharpe ratio as P and the same volatility as the market portfolio.
- (b) Show that if you compare $E[r_{P^*}^e]$ and $E[r_M^e]$, you get the same ranking as if you compare the Sharpe ratio of P and M . Explain why the second comparison could be more intuitive than the first one
- (c) How would you optimally combine portfolio P and M given that portfolio P has a market beta of 1.2? *Hint: use a single-factor model to answer this question.*
- (d) What is the Sharpe ratio of the optimal portfolio?

Numerical Exercise (in Python)

You are asked to evaluate the performance of four funds: T Rowe Price Blue Chip Growth (Ticker: TRBCX), Goldman Sachs Large Cap Growth Insights (Ticker: GLCGX), BlackRock Exchange Portfolio BlackRock Shares (STSEX), and DFA U.S. Small Cap Value Portfolio (Ticker: DFSVX) over the period from January 2002 through July 2026 (last date 2026-07-01).

1. Load monthly price data from Yahoo Finance for each fund and the risk-free rate and FFC 4-factor returns from the Kenneth French data library. Compute monthly excess returns. *Hint: For fund returns, start in December 2001 so that you can compute returns from January 2002 onward.*
2. Report the Sharpe ratio and the Information ratio (relative to the market) over the period for each fund.
3. Can you find evidence of fund manager skills relative to the FF 3-factor model? How about the FFC 4-factor model? *You should use a regression analysis to answer this question.*